



Final Project Report

VitaSync: A Wearable Multi-Sensor System for Early Heat Stroke Risk Detection

BM2210: Biomedical Device Design
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1 Introduction

Team VitaSync recognized the need to constantly monitor when people stop sweating during extreme heat or heavy exercise to prevent severe heatstroke before it causes serious health problems. This report details the final design and testing of our solution: **a Smart Vest** equipped with sensors to measure sweat levels, heart rate, skin temperature, and the surrounding environment. The vest sends all this real-time data to a connected mobile application, where a machine learning model continuously analyzes the information to make smart decisions and instantly alert the user if they are at risk.

1.1 Problem Statement

A low-cost, non-invasive wearable is required that can fuse galvanic skin response, environmental heat index, PPG, and skin temperature data to flag the earliest signs of heat stroke risk in real time, without requiring clinical supervision.

2 System Overview

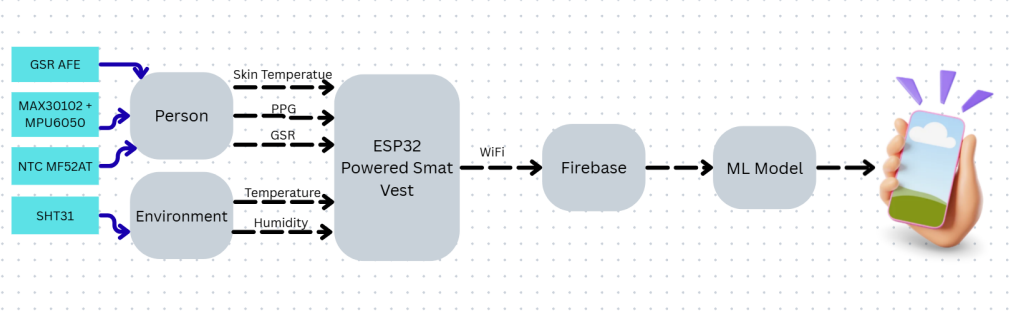


Figure 1: High-level system block diagram

The system consists of four functional layers:

(a) **Sensing layer** – GSR analog front end, MAX30102 PPG sensor, MF52AT thermistors, SHT31 ambient sensor.

(b) **Processing layer** – ESP32-WROOM-32 microcontroller performing signal conditioning, lock-in demodulation, sensor fusion, local threshold alerting, and edge-based ML model inference.

(c) **Communication layer** – Wi-Fi telemetry to a companion mobile application and, optionally, a cloud backend.

(d) **Application layer** – mobile app for individual users and a fleet-monitoring web dashboard for the B2B occupational safety deployment.

3 Component Selection

ESP32 Microcontroller: Handles processing, edge ML inference, and signal conditioning. It provides Wi-Fi and Bluetooth for telemetry. Its internal DAC generates AC waves for the GSR circuit.

MAX30102 PPG Sensor: This is a fully integrated optical module. It reduces noise and saves board space. It captures continuous heart rate data at the C7 vertebra.

MPU6050 Accelerometer: Tracks user movement and vibration. It is mounted exactly beside the PPG sensor. The system uses it to discard motion-distorted heart rate readings.

MF52AT NTC Thermistors: These offer high sensitivity and fast response times. They are placed directly on the skin. They track rapid skin temperature spikes.

SHT31 Environmental Sensor: Measures ambient temperature and humidity accurately. It is placed on the outer shoulder. This strictly isolates weather data from trapped body heat.

OPA2333 Operational Amplifier: A precision, zero-drift amplifier. It maintains a stable 1.65V virtual ground. It converts minute sweat microcurrents into measurable voltages.

74HC4066 Analog Switch: Routes AC signals between a calibration resistor and the skin. This enables real-time system self-calibration. It ensures accurate baseline impedance readings.

GSR Surface Electrodes: Placed 2.0 to 3.0 cm apart on the upper back. They measure electrodermal activity to detect sweat cessation. This spacing prevents short-circuits from merging sweat droplets.

4 GSR Circuit Design and Explanation

The galvanic skin response (GSR) measurement sub-system is an AC-excited analog front end built around three OPA2333 (dual, zero-drift, micro-power) operational amplifiers, operating from a single 3.3 V supply shared with the ESP32.

4.1 Design Rationale: AC Excitation vs. DC Excitation

A DC excitation current through the skin causes electrode polarization, skin irritation, and long-term measurement drift. So this instead injects a small AC current at a known frequency and measures the resulting voltage using synchronous (lock-in) demodulation, which rejects DC offsets, 50/60 Hz mains interference, and 1/f noise.

4.2 1.65 V Virtual Ground Reference

Because the AFE runs from a single 3.3 V rail, a mid-rail reference is required so AC signals can swing symmetrically without clipping against ground.

$$V_{\mathrm{REF}} = \frac{V_{CC}}{2} = \frac{3.3\,\mathrm{V}}{2} = 1.65\,\mathrm{V} \tag{1}$$

This is generated using a 1 MΩ/1 MΩ resistive divider, a 1 μF decoupling capacitor at the divider node, and a unity-gain op-amp buffer (voltage follower) to provide a low-impedance source for the rest of the circuit.

4.3 Excitation Signal Generation

A sinusoidal excitation signal is synthesized digitally by the ESP32's onboard 8-bit DAC (GPIO25) and smoothed by an active second order Sallen Key low pass filter (unity gain, Butterworth aligned) to remove the staircase quantization artifacts of the DAC output.

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \tag{2}$$

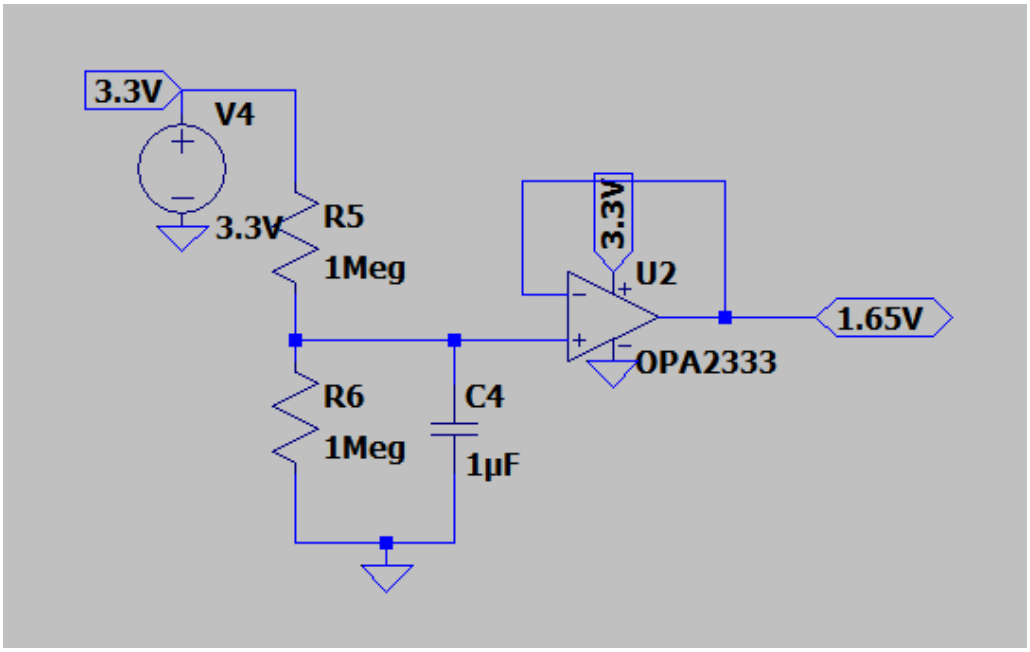


Figure 2: Virtual ground reference generator schematic

With $R_1 = R_2 = 10\,\text{k}\Omega$, $C_1 = 1\,\text{nF}$, $C_2 = 2.2\,\text{nF}$:

$$f_c = \frac{1}{2\pi\sqrt{(10{,}000)^2 \times (1 \times 10^{-9})(2.2 \times 10^{-9})}} \approx 10.7\,\text{kHz} \tag{3}$$

This cutoff is comfortably above both excitation frequencies used (125 Hz and 1 kHz), so the passband is flat at the frequencies of interest while DAC quantization images (which appear at multiples of the 20 kHz DAC update rate) are attenuated.

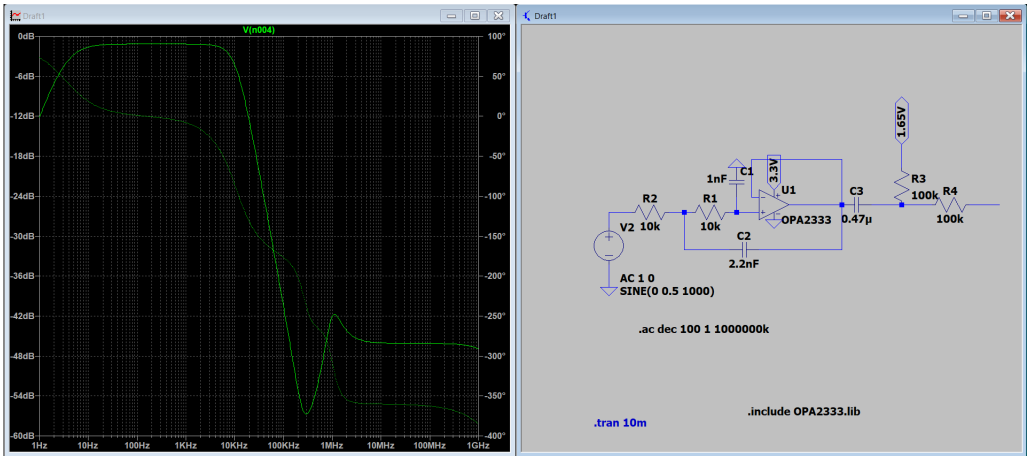


Figure 3: Sallen Key low pass filter schematic and simulated frequency response

4.4 AC Coupling and DC Biasing

A 0.47 μF series capacitor blocks any residual DC component from reaching the skin, and a 100 kΩ resistor re-biases the AC signal around the 1.65 V virtual ground. The resulting high-pass corner is:

$$f_{hp} = \frac{1}{2\pi RC} = \frac{1}{2\pi(100{,}000)(0.47 \times 10^{-6})} \approx 3.4\,\text{Hz} \tag{4}$$

which is far below both excitation frequencies, so the excitation passes essentially unattenuated while DC is fully blocked. A series 100 kΩ resistor (R_s) limits the maximum current that can flow through the skin to a safe microampere-level value.

4.5 Analog Path Switching

A 74HC4066 quad bilateral analog switch IC alternates the excitation path between a known 10 kΩ, 0.1% precision reference resistor (SW1) and the skin electrodes (SW2), controlled by ESP32 GPIO16 and GPIO17 respectively. The two switches are never enabled simultaneously; a software-enforced dead time of approximately 2 ms is applied between transitions.

4.6 Transimpedance Amplifier (TIA)

The current returning through the selected path is converted to a voltage by an inverting transimpedance amplifier:

$$V_{\text{out}} = -I \times Z_f, \qquad Z_f = \frac{R_f}{1 + j\omega R_f C_f} \tag{5}$$

with $R_f = 100\,\text{k}\Omega$ and $C_f \approx 220$–$330\,\text{pF}$ (C0G/NP0 dielectric, selected for stability and low temperature drift).

4.7 Ratiometric Resistance Calculation

Because absolute excitation amplitude, op-amp gain, and ADC reference voltage all carry some uncertainty, the skin resistance is computed as a *ratio* relative to the known reference resistor, canceling most systematic errors:

$$R_{\text{skin}} = (R_s + R_{on} + R_{\text{ref}}) \times \frac{A_{\text{ref}}}{A_{\text{skin}}} - R_s - R_{on} \tag{6}$$

where A_{ref} and A_{skin} are the demodulated signal amplitudes measured on the reference and skin paths respectively, and R_{on} is the 74HC4066 on resistance ($\approx 120\,\Omega$ typical). A "bracketed" measurement sequence (Reference → Skin → Reference) is used, with the two reference readings averaged, to cancel slow drift in the DAC/ADC clock and excitation amplitude between measurements.

4.8 ADC Interface

The TIA output is buffered and passed through an RC anti-alias filter (1 kΩ + 10 nF) into the ESP32's ADC1 channel (GPIO32). ADC1 is used specifically because ADC2 is unreliable while the Wi-Fi radio is active on the ESP32.

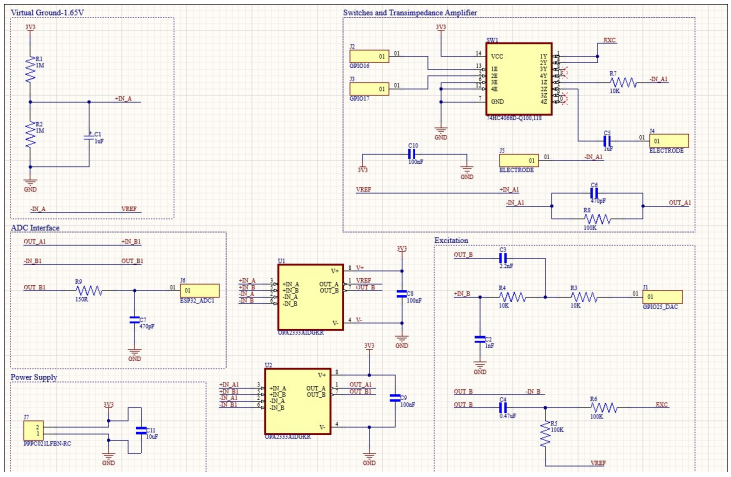


Figure 4: Full schematic

5 Sensor Placement and Wearable Ergonomics

Anatomical placement was selected based on a combination of physiological signal quality, motion stability, and wearability feedback gathered from athletes during ideation.

Table 1: Final sensor placement summary and justification

Sensor	**Location**	**Justification**
MAX30102 (PPG) + MPU6050	Base of neck (C7 zone), skin-facing	Good perfusion near major vessels, relatively low motion from breathing/arm movement compared to chest wall, co-located accelerometer enables motion-artifact cancellation referenced to the sensor's own movement.
GSR electrodes	Upper back, between the scapulae (trapezius), exterior-facing housing	Highest thermoregulatory (heat-induced) sweat gland density region; more mechanically stable than the front during arm movement; physically separated from PPG electronics to reduce electrical interference.
Thermistors (x2)	Bilateral upper pectoral (chest), symmetric left/right	Muscle tissue (not bone) gives better perfusion and faster thermal response than spinal/scapular bone placement; symmetric placement allows averaging to reduce single-side strap or posture artifacts; physically separated from the GSR electrodes to avoid evaporative-cooling bias.
SHT31 (ambient temperature/humidity)	Outer-facing shoulder, standoff from fabric	Measures true ambient heat index rather than body-generated microclimate; standoff mounting avoids trapped body heat skewing the reading.

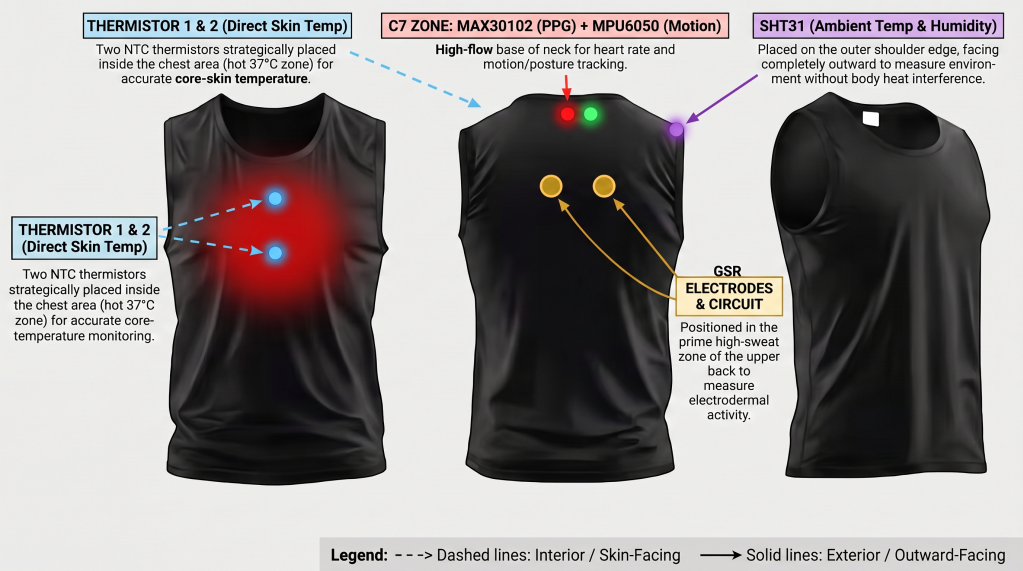


Figure 5: Final sensor placement diagram on vest (front and back views)

5.1 Rejected Placements

Forehead, ear canal, and armpit placements were considered and rejected during concept screening: sweat flooding corrupts optical/resistive readings at the forehead and armpit, and the ear canal cannot accommodate a multi-sensor array at the required accuracy and comfort level.

6 Firmware Design

6.1 Microcontroller and Pin Mapping

The system firmware runs on an ESP32-WROOM-32 (dual-core, Wi-Fi/BLE). The hardware components are connected to the ESP32 using its available communication interfaces and analog input pins. The pin assignments are defined in Config.h, making it easier to manage the hardware configuration. Table 2 lists the confirmed pin assignments for a standard 30-pin DevKitV1 board.

Table 2: ESP32 pin assignments

Signal	GPIO	Function
DAC excitation output	GPIO25	Sallen-Key filter input (DAC1)
SW1 control	GPIO16	74HC4066 reference-path select
SW2 control	GPIO17	74HC4066 skin-path select
ADC input	GPIO32	TIA/buffer output (ADC1_CH4)
ADC input	GPIO34	NTC Thermistor_1 (ADC1_CH6)
ADC input	GPIO35	NTC Thermistor_2 (ADC1_CH7)
I2C SDA	GPIO21	MAX30102, SHT30D, MPU6050
I2C SCL	GPIO22	MAX30102, SHT30D, MPU6050
Buzzer control	GPIO27	Alarm buzzer output
Onboard LED Control	GPIO02	Critical Alert through LED

6.2 Overall Firmware Architecture and ESP32 Utilization

The firmware is developed in C++ using the PlatformIO development environment. It takes advantage of the ESP32's dual-core architecture and FreeRTOS capabilities to manage multiple tasks simultaneously. This allows the system to collect sensor data continuously while handling more demanding operations, such as digital signal processing, machine learning-related calculations, and network communication.

The firmware is organized into separate modules, each responsible for a specific function. Hardware-related components are handled by libraries such as `Max30102.h`, `Mpu6050.h`, and `Sht3x.h`, while data processing and communication are managed by modules such as `PpgProcessor.h`, `HeatMetrics.h`, and `FirebaseUploader.h`. This modular structure makes the code easier to maintain and helps keep the main program responsive.

The ESP32 uses separate FreeRTOS tasks or asynchronous processing techniques to manage sensor readings and network communication independently. For example, the system can continuously collect PPG data from the MAX30102 without being significantly affected by delays caused by Wi-Fi communication or Firebase uploads.

6.3 Sensor Logic and Data Processing Pipeline

The system follows a step-by-step process in which sensor data is collected, filtered, analyzed, and used to estimate the user's heat-related risk. The overall workflow is as follows.

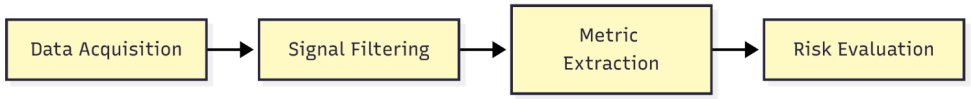


Figure 6: The overall workflow

- **PPG and Heart Rate:**
  - The MAX30102 collects raw photoplethysmography (PPG) signals from the user's skin. However, these signals can be affected by movement, background noise, and changes in sensor contact. To improve signal quality, the raw data is processed using filters defined in `Filters.h`, which help reduce baseline fluctuations and unwanted high-frequency noise.
  - After filtering, the `PpgProcessor.h` and `SpectralHr.h` modules analyze the signal using the techniques, peak detection and frequency-domain analysis. These methods help identify pulse patterns and estimate the user's heart rate more reliably.
- **Motion and Activity Monitoring:**
  - The MPU6050 continuously measures the user's acceleration and rotational movement. The MotionDetector module analyzes these readings to identify activity levels, such as resting or active movement. It can also help identify periods when motion may interfere with PPG measurements, allowing the system to account for potentially unreliable readings. Depending on the implemented detection logic, the motion data may also be used to support fall detection.
- **Environmental and Skin Temperature Monitoring:**
  - The SHT30D sensor measures the surrounding temperature and relative humidity, providing information about the environmental conditions the user is exposed to. At the same time, the NTC thermistor measures skin temperature, helping the system

monitor changes in the body's surface temperature. Together, these measurements provide useful information for assessing possible heat stress.

- **Sweat and Skin Conductance Monitoring (GSR):**
  - The firmware manages Galvanic Skin Response (GSR) data collection by first using a Digital-to-Analog Converter (DAC) to provide a stable excitation voltage to the analog front end. It then continuously reads the resulting voltage changes through the Analog-to-Digital Converter (ADC) and converts them into digital values. These changes are used to monitor variations in skin conductance, which can occur as sweat gland activity changes.

    After collecting the data, the raw signal is processed using a two-stage digital filtering process. First, a low-pass filter, operating within the frequency range from 1 Hz to 5 Hz, helps reduce high frequency electrical noise and unwanted signal variations caused by movement. Next, a high pass filter is used to separate the slower changes in the signal from its faster fluctuations. This allows the system to identify two main components: the tonic component, which represents gradual changes in skin conductance, and the phasic component, which captures short-term responses associated with sudden increases in sweat gland activity.

6.4 Heat Stroke Risk Prediction Logic

The main decision-making process is handled by the HeatMetrics and HeatRisk modules. Rather than depending on a single sensor or a fixed threshold, the system combines environmental and physiological measurements to estimate the user's level of heat-related risk. The prediction process involves three main stages:

1. Environmental Heat Assessment: The temperature and humidity readings collected by the SHT30D sensor are used to estimate the level of environmental heat exposure.

2. Physiological Strain Assessment: The system combines skin temperature readings from the NTC thermistor, heart rate estimates from the PPG processing modules, and changes in skin conductance from the GSR sensor. For example, an increase in skin temperature together with an unusually elevated heart rate and changes in skin conductance may indicate that the user is experiencing increased physiological strain. These measurements are interpreted alongside activity levels and environmental conditions to provide a more complete assessment.

3. Risk Classification: The `HeatRisk.h` module evaluates the processed metrics and assigns the user to a risk category, such as Safe, Warning, or Critical. This classification allows the system to identify potentially concerning conditions and provide an early indication when the user's heat-related risk increases.

By combining multiple sources of information, the system aims to provide a more reliable assessment than one based on temperature or heart rate alone. However, the resulting categories represent estimated risk levels rather than a confirmed medical diagnosis of heat stroke.

6.5 Network Connectivity and Telemetry

After processing the sensor readings and calculating the heat-risk metrics, the system uses the FirebaseUploader module to transmit the collected information to a remote platform. This makes it possible to monitor the user's condition without needing to access the wearable device directly.

- **Wi-Fi Connectivity:** The ESP32 connects to a Wi-Fi network to communicate with the remote server. The firmware includes connection monitoring and automatic reconnection mechanisms to restore communication if the connection is interrupted.
- **Firebase Integration:** The `FirebaseUploader` module organizes the collected information into structured data (JSON payloads), containing timestamps, physiological measurements, environmental readings, and calculated heat-risk levels. These records are then uploaded to Google Firebase to integrate with the mobile application.

7 Data Propagation Architecture

7.1 How Data Flows

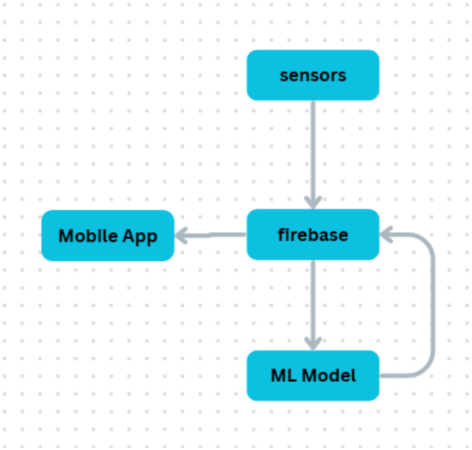


Figure 7: How data Flows

The system follows a continuous real-time data flow pipeline:

1. **Sensors to Cloud:** The wearable sensors collect physical health metrics (such as skin temperature, heart rate, and sweat levels) and send them to Firebase via an onboard microcontroller (ESP32) over Wi-Fi.
2. **Machine Learning Anomaly Detection:** The cloud hosted ML Model constantly reads the incoming sensor data from Firebase, evaluates risk patterns (like heat stress triggers), and writes the anomaly status back into Firebase.
3. **App Synchronization:** The Mobile App continuously listens to Firebase in real time, updating the UI gauges instantly and triggering high priority safety warnings if an anomaly is flagged.

7.2 Firebase Data Structure

Field Breakdown:

- `devices/DEV-001`: The top-level root node mapping each physical wearable device to its specific ID.
- `feels_like_temp`: Calculated heat index combining temperature and environmental factors (e.g., 35.5°C).
- `heart_rate`: Real-time heart rate measurement in beats per minute (bpm) (e.g., *78bpm*).

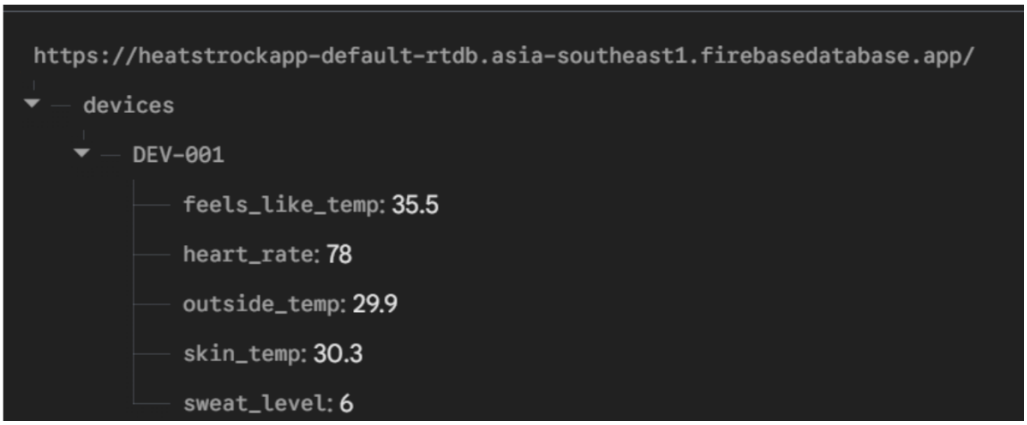


Figure 8: Firebase Data Structure

- `outside_temp`: Ambient environmental temperature surrounding the worker (e.g., 29.9°C).
- `skin_temp`: Direct skin surface body temperature measured by the vest sensors (e.g., 30.3°C).
- `sweat_level`: Galvanic Skin Response (GSR) intensity value tracking perspiration (e.g., 6).

8 Mobile Application

8.1 Purpose of the Heat stroke warning Mobile Application

The Heat stroke warning mobile application is designed to continuously stream real time physiological and environmental data such as body temperature, heart rate, and sweat levels directly to workers and management. By utilizing cloud hosted machine learning (Isolation Forest), the app detects precise anomaly patterns (such as sudden spikes in skin temperature combined with a dangerous drop in sweating) and immediately pinpoints the exact physiological reason for a heat stress warning. Industrially, this allows companies to monitor employee health conditions in real time across job sites, significantly reducing workplace risks and guaranteeing high safety standards for workers before critical physiological collapse occurs.

8.2 Language and Framework Selection

The VitaSync mobile application is built using Flutter, an open source UI software development kit created by Google.

- **Why Flutter?**
  - Flutter uses Dart as its programming language. It allows developers to compile native mobile applications for both Android and iOS from a single codebase. For this project, Flutter provides high performance rendering, smooth UI animations for vitals monitoring (such as gauge progress bars and wave effects), and rich asynchronous support required for real time data streaming

8.3 Summary of Building the Heat stroke warning App

- **Project Initialization:** The process begins by creating a new Flutter project structure using terminal commands.

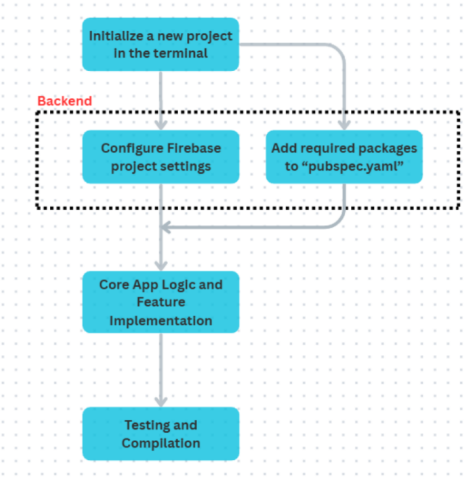


Figure 9: Architecture of the app

- **Backend Configuration:** The application setup links to backend services by configuring Firebase settings and adding essential dependency packages into the `pubspec.yaml` configuration file.
- **Core App Logic and Feature Implementation:** Key application features are built, including real-time telemetry streaming from Firebase, animated health gauge widgets, and high-priority heat stroke alert notification systems.
- **Core App Logic and Feature Implementation:** Key application features are built, including real-time telemetry streaming from Firebase, animated health gauge widgets, and high-priority heat stroke alert notification systems.
- **Testing and Compilation:** Finally, the app is debugged on connected devices and compiled into a deployable release APK.

8.4 Safety Thresholds Used in the System

To provide reliable early warnings before physical collapse occurs, this App monitors both static threshold limits and dynamic ML anomalies:

- **Skin Temperature Threshold:** 38.5°C (indicates core body heat retention).
- **Heart Rate Exertion Threshold:** >130 to 140bpm during prolonged outdoor exposure.
- **Environmental Heat Index Threshold:** 40°C combined humidity index.
- **GSR / Sweat Anomaly:** Sudden flatlining or abnormal drop in sweat conductivity while skin temperature continues to rise (Anhidrosis warning).

8.5 Critical Components and Implementation Details

The mobile application handles several critical real time features to ensure user safety:

- **Firebase Realtime Database Integration:** The app connects directly to Firebase to continuously listen for streamed health and environmental data (`skin_temp, heart_rate, sweat_level`, and anomaly detection flags).

- **Real Time Stream Listeners:** Utilizing `StreamBuilder`, the UI instantly updates whenever new data packets arrive from the hardware device, eliminating the need for manual page refreshes.
- **Multi Tier Alert System & State Management:** The app triggers warning popups and auditory alerts based on risk levels. Critical anomalies trigger high priority modal dialogs that lock user attention until acknowledged.
- **Background Permissions:** Integrates local notification plugins and permission handlers to ensure alerts bypass silent modes during dangerous heat stroke conditions.

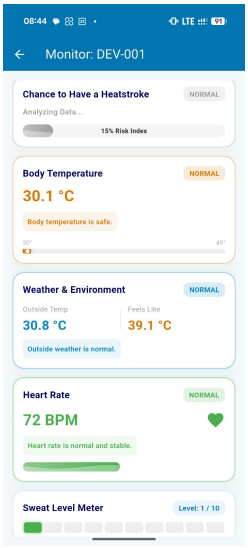


Figure 10: User interface of mobile application

9 Machine Learning Model for Risk Classification

9.1 Purpose

The firmware's rule-based score (NORMAL → CAUTION → WARNING → ALERT) uses fixed thresholds that are the same for everyone. The ML model adds a personalised risk estimate. It learns how each wearer's signals change together over time, instead of checking each signal against its own threshold. This directly answers the Medical Officer's point that a single indicator is not reliable.

9.2 Data Pipeline

- ESP32 → Firebase Realtime DB (`devices/<ID>`) → Python service → risk level written back to Firebase → Flutter app shows it and sends the notification.
- The model reads from Firebase, not the sensors, so it can be retrained without changing the firmware.
- The local buzzer stays rule-based. It still works if there's no Wi-Fi or the model is down.

9.3 Input Features

Computed over sliding windows, e.g. 1–5 minm Trends and baseline-relative values matter more than raw readings, because heat stroke shows up as a pattern: temperature and HR rising while sweat falls.

Table 3: Feature groups used for heatstroke risk estimation

Group	Features
Body	Skin temperature, rate of rise (°C per 10 min)
Cardiac	Heart rate, HR as % of age-predicted max ($HR_{max} = 208 - 0.7 \times age$), HR trend
Sweat	Sweat level (1–10), change from the wearer's own baseline, drop from recent peak (anhidrosis sign)
Environment	Outside temperature, humidity, feels-like temperature (heat index)
Context	Activity level from the MPU6050, signal quality flags (HR confidence, electrode contact)

9.4 Labels

- Real heat-stroke events cannot be recorded ethically, so the model needs a proxy target. We use the Physiological Strain Index (PSI).

$$\mathrm{PSI} = 5\,\frac{T_t - T_0}{39.5 - T_0} + 5\,\frac{\mathrm{HR}_t - \mathrm{HR}_0}{180 - \mathrm{HR}_0} \tag{7}$$

- The scale is 0–10. It is binned into three classes: Low (< 4), Moderate (4–7) and High (> 7).
- Alternatives:
  - Use the firmware risk-engine levels as labels to start with.
  - Add synthetic sweat-failure segments, matching the injected GSR test in the feasibility plan.
- T_t is skin temperature, a proxy for core temperature, calibrated against a clinical thermometer.

9.5 Model Choice

- Main model: Random Forest or gradient-boosted trees. They work well on small tabular datasets, are fast, and show which features drive the prediction.
- Comparison baseline: logistic regression and the existing rule-based score.
- Output: risk class plus a probability, which lets the app give graded alerts (medium vs critical notification channels already exist).

9.6 Evaluation

- Leave-one-subject-out cross-validation. Never test on a person the model was trained on.
- Main metric is recall (sensitivity) of the High-risk class, because a missed alert is far worse than a false one. Also report precision, F1, and the confusion matrix.
- Report lead time: how many minutes before the High threshold the model raises a warning.
- Show feature importance to check the model relies on physiologically sensible inputs.

9.7 Limitations

- Small dataset from a few healthy subjects.
- Proxy labels instead of clinical events.
- Skin temperature is not core temperature.
- The model screens for risk; it does not diagnose.

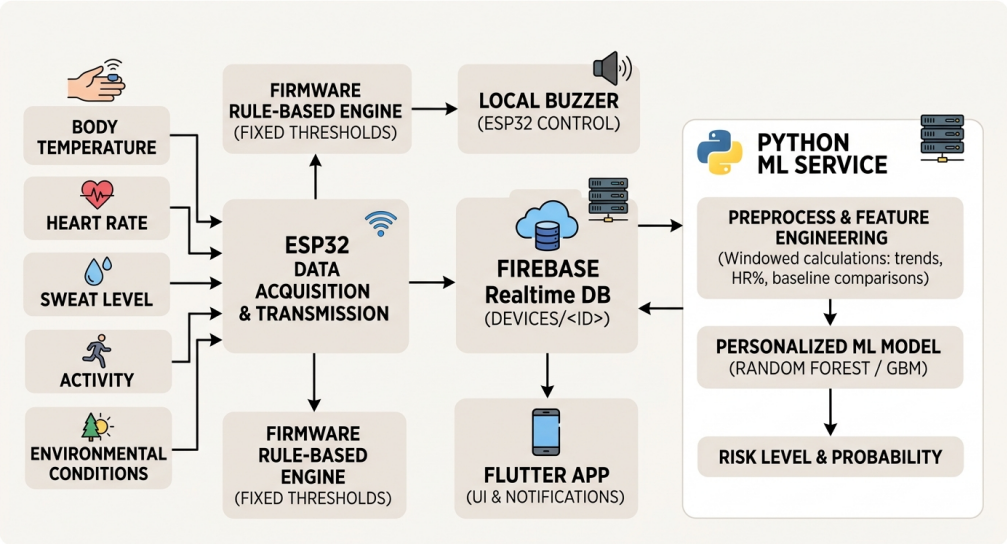


Figure 11: End-to-end data flow: sensors, ESP32, Firebase, Python ML service and Flutter app. The local buzzer stays rule-based and works without Wi-Fi.

10 Validation and Testing

10.1 Analog Front-End Bring-Up Validation

A self-test routine measures the known 10 kΩ, 0.1% reference resistor on power-up before any skin measurement is attempted. A reading outside a ±15% window halts the device with a diagnostic message rather than reporting unvalidated skin data. This process surfaced and led to correction of two issues during development: an `int32_t` accumulator overflow in the lock-in algorithm, and a capacitor value mix-up on the AC-coupling stage (an incorrectly substituted 1 nF capacitor in place of 0.47 μF, which shifted the intended high-pass corner from ≈ 3.4 Hz up into the kilohertz range and blocked the intended excitation frequencies).

10.2 Switching Sub-System Verification

The 74HC4066 switching stage was independently verified using bench equipment: logic-level check (multimeter), on-resistance measurement (multimeter, known series resistor method), two-channel oscilloscope verification of SW1/SW2 dead-time (no simultaneous activation), signal-integrity comparison across the switch, and off-state isolation/leakage testing.

10.3 Filter Verification

The Sallen-Key low-pass filter's frequency response was verified both in LTspice simulation and on the physical breadboard using a standalone DAC sine-wave generator sketch and an oscilloscope, confirming the expected $\approx 10.7\,\text{kHz}$ cutoff and flat passband across both excitation frequencies.

10.4 Benchtop Sensor Calibration and Cross-Talk Verification

SHT31 accuracy is validated in a controlled environmental chamber across 25–45 °C (target accuracy $\pm 0.2\,^{\circ}\text{C}$), with synthetic GSR signals injected to confirm software suppression of false-positive alarms caused by isolated-marker anomalies (e.g. salt-crystal conductivity after true anhidrosis onset).

10.5 Motion Artifact and Optical Shielding Stress Test

PPG tracking stability is validated during dynamic athletic movement under direct sunlight, compared against a reference chest-strap heart-rate monitor (target accuracy $\pm 4\,\text{bpm}$), with zero optical saturation from ambient light leakage.

10.6 In-Situ Field Validation

Full-system endurance (> 4 hours) and real-time alarm delivery are validated during supervised heat-acclimation training sessions, with simulated threshold breaches used to confirm local buzzer actuation and mobile push-notification delivery within < 2 seconds, without skin irritation or motion discomfort over the session duration.

Table 4: Summary of validation tests and pass/fail criteria

Test	**Criteria**	**Status**
Reference resistor self-test	Within $\pm 15\%$ of $10\,\text{k}\Omega$	*INSERT RESULT*
Sallen-Key cutoff	$10.7\,\text{kHz} \pm 10\%$	*INSERT RESULT*
PPG under motion	$\pm 4\,\text{bpm}$ vs. reference strap	*INSERT RESULT*
Alarm latency	$< 2\,\text{s}$	*INSERT RESULT*

11 Industrial-Level Scaling

To move VitaSync from a single prototype to large-scale deployment across construction sites, estates, factories and heavy-industry facilities, the system is designed to scale in four areas: manufacturing, cloud backend, fleet management and firmware maintenance.

11.1 Manufacturing Transition

The breadboard/dot-board prototype (discrete OPA2333 op-amps, 74HC4066, ESP32-WROOM-32 module) translates directly to a custom PCB for volume manufacturing, with the following considerations:

- Migration from through-hole prototyping to a 2–4 layer PCB with dedicated analog and digital ground planes connected at a single star point.
- Component sourcing in tape-and-reel quantities for SMT assembly through a contract manufacturer (e.g. a local or regional EMS provider).
- Design-for-manufacturing (DFM) review of the enclosure, moving from prototype FDM/SLA printing to injection-molded TPU housings at scale.
- Modular sensor harnesses (I2C/SPI buses), so a faulty sensor can be replaced quickly and cheaply on site without replacing the whole vest.

11.2 Scalable Cloud Architecture

- **Per-device data nodes:** the Firebase Realtime Database indexes each vest by a unique serial number (`DEV-001`, `DEV-002`, ..., `DEV-N`) under the top-level `devices` tree. Each wearable writes only to its own node, so thousands of vests can stream telemetry at the same time without overwriting each other's data.
- **Parallel ML processing:** the Python ML service handles incoming data in parallel batches, so risk classification can run across a large workforce in near real time. The service can be scaled out with more workers as the fleet grows.

11.3 Multi-User Hierarchy and Fleet Management

The system serves two levels of users.

- **Worker level:** each worker sees their own live vitals in the mobile app and receives personal heat-stress warnings and audio alerts.
- **Supervisor level:** safety officers and site managers use a centralized web/mobile fleet dashboard that aggregates data from all active vests on site. Workers are grouped by risk status (Normal, Caution, Critical), so supervisors can intervene across the whole site at once.

11.4 Firmware Maintenance

Over-the-air (OTA) updates let security patches, sensor calibration updates and algorithm improvements be pushed wirelessly to hundreds of deployed vests at once, without collecting the devices.

11.5 Supply Chain and Quality Control

The following quality gates are recommended before shipment:

- incoming component inspection (op-amp offset/bias verification, resistor tolerance batch sampling);
- in-circuit test (ICT) fixtures running the GSR AFE self-test routine (Section 10) at the end of the production line;
- burn-in testing.

11.6 Deployment Model

Under the Hardware-as-a-Service model (Section 15), industrial clients lease vest fleets and scale them up or down with seasonal workforce size, paying a monthly per-worker subscription for the fleet dashboard.

11.7 Final Product Fabrication Plan

1. **Main body**
   - Recycled polyester (88%) / elastane (12%) micro-mesh knit
   - About 140 gsm, 4-way stretch
   - Moisture-wicking, quick-dry, UPF 50+
2. **C7 sensor zone (PPG + MPU6050)**
   - Flexible TPU pod (Shore 85A), matte black
   - 3 mm polyester 3D spacer mesh base for airflow
   - Matte-black silicone gasket to block ambient light
   - Optically clear silicone window over the PPG sensor
   - Bonded powernet anti-slip patch with silicone grip dots
   - Brushed microfibre skin liner, hypoallergenic
3. **GSR electrode zone (upper back)**
   - Two silver-plated nylon conductive-knit pads, 2–3 cm apart
   - TPU-laminated insulating border around the pads
4. **Chest thermistor pockets**
   - Sealed TPU film micro-pockets, left and right
   - Microfibre lining so the sensor touches skin, not seams
5. **Ambient sensor mount (SHT31, outer shoulder)**
   - Raised cap standing off the fabric
   - PTFE hydrophobic vent
   - Reflective ripstop top
   - Keeps body heat out of the ambient reading
6. **Side ventilation panels**
   - Laser-perforated power mesh from armpit to hem
   - Gives airflow without losing fit
7. **Battery pouch (low back)**
   - Sealed TPU case
   - Closed-cell foam lining
   - Snap-out electronics module for washing
8. **Finishing and wiring**

- Flatlock stitching and bonded hems, no chafe at neck or arms
- Insulated silicone leads in bonded fabric channels, with strain relief
- Buzzer behind an acoustic mesh window at the collar
- Segmented retroreflective hi-vis tape on the lower torso
- Care: remove the module, wash cold, air dry



Figure 12: Fabric Spotlight

12 Stakeholder Analysis

Table 5: Key stakeholders, their roles, needs and influence

Stakeholder	**Role**	**Main interest / needs**	**Influence**
Outdoor workers (construction, agriculture, tea/rubber estates)	Primary users	Comfort over a full shift, no distraction, alerts they trust	High (adoption)
Athletes, cricketers in day matches	Primary users	Thin, ventilated, no moving parts; works while running	Medium
Military personnel	Primary users	Rugged, reliable in extreme heat	Medium
Elderly people in heatwaves	Primary users	Simple use, caregiver alerts	Medium
OHS managers, site supervisors	Buyers / alert receivers	Fleet dashboard, fewer incidents, less liability	High (purchase decision)

Continued on next page

Table 5 (continued)

Stakeholder	Role	Main interest / needs	Influence
Company executives, military procurement, elderly-care administrators	Decision makers	Cost per worker, ROI, scalability	High
Medical officers (consulted: Dr. J.D. Fernando, NTH Kandy)	Clinical advisors	Multi-parameter monitoring, low false negatives	High (design input; led to adding HR)
Insurance companies	Secondary	Fewer heat-injury compensation claims	Medium
Labour unions	Secondary	Worker safety, data privacy	Medium
Regulators (NMRA, FDA)	Regulatory	Product claims and classification	High
Team VitaSync	Developers	Feasible within one semester and a student budget	—

Key takeaways:

- **Users and buyers are different people.** The worker wears it; the employer pays. This is why the B2B occupational-safety model (Hardware-as-a-Service) was chosen.
- **Clinical input changed the design.** The doctor's single-indicator concern added PPG heart rate. Athletes' feedback set the form factor (thin, ventilated, upper-back/C7 placement).
- **Regulation shapes the wording.** The product is positioned as general-wellness risk monitoring. Claiming to diagnose or prevent heat stroke would make it a Class II device needing clinical trials and ISO 13485.
- **Privacy concern.** Employers will see workers' physiological data, which matters to unions and workers. Access control and consent are needed.

13 Intellectual Property Considerations

13.1 Prior Art

Heat-stroke detection using multi-parameter models is not novel in the biomedical device sector, and several systems with similar architectures already exist. Notable examples include:

1. A heat-stroke warning system for athletes: https://pmc.ncbi.nlm.nih.gov/articles/PMC11769488/
2. HeatGuard, an arm patch for heat-stroke detection: https://www.mdpi.com/1424-8220/26/8/2556

Most of these research-level solutions combine GSR sensors, thermistors and PPG, and pass the readings to a machine learning model. VitaSync differentiates itself by acquiring physiological data from multiple optimized points across the body, integrating a custom machine learning model, and embedding the hardware in a cohesive, easy-to-use smart vest.

13.2 Existing Patents

Table 6 lists previous patents with similar aims.

Table 6: Previous similar patents

Patent	Link	Summary
US9547972 B2 – Methods and systems for emergency alerts	https://patents.google.com/patent/US9547972	Alert system using GSR to identify heat stroke and heat stress.
US9701243 B1 – Heatstroke safety system	https://patents.google.com/patent/US9701243	Uses a motion sensor, a location sensor and a temperature sensor to monitor heat-stroke risk.
US8087254 B2 – Personal heat control device and method	https://patents.google.com/patent/US8087254B2	Wearable helmet that detects heat stroke and heat stress and provides cooling.
US8434163 B1 – Vest system that provides for a targeted heat/cool therapy treatment	https://patents.google.com/patent/US8434163	Therapy vest with shoulder, side and back pockets that hold heat or cold packs for targeted body-temperature therapy.
US11957188 B2 – Wearable device, and sensor device	https://patents.google.com/patent/US11957188	Clothing-attached sensor housing with a radiant heat sensor on the front and a temperature/humidity sensor on the back, designed to measure humidity inside clothing accurately.
JP5022506 B1 – Heat stroke detection system	https://patents.google.com/patent/JP5022506B1	System that can be installed into existing protective wearable equipment, such as radiation protection equipment.

13.3 Patenting Strategy

1. **Component-based patenting.**
   (a) The GSR circuit and multi-zone sensor placement are filed together as one patent. This can include a future breathable garment with a distinctive material composition for the main garment and the C7 area where the sensors sit.
   (b) The ML model and other software are difficult to patent, so the model is kept as a trade secret rather than patented.
2. **Patent landscaping and claim-by-claim comparison.** Map the existing patents in the field, break our solution down feature by feature, and compare each feature against the claims of those patents.
3. **Freedom to Operate (FTO) search and implementation** (see below).

13.4 Filing Process

Local filing (NIPO, Sri Lanka).

1. File an application on Form P1 with the National Intellectual Property Office (NIPO), with a full description of the invention, the claims and the required fee.

2. Submit a search report, either from an International Searching Authority or requested through the local NIPO examiners, to verify novelty.
3. Substantive examination of novelty, inventive step and industrial utility.
4. Publication in the Government Gazette.
5. Grant of the patent, valid for 20 years from the filing date, subject to annual renewal fees from the second year.

International route. A Patent Cooperation Treaty (PCT) application gives a single filing and priority date, and the applicant has up to 30 months from that date to commit to filing in individual countries. NIPO cannot act as the PCT receiving office, so the filing goes through WIPO. The key target countries are the USA, Japan, Taiwan, South Korea and China.

13.5 Hurdles to Overcome

- **Existing research:** GSR, temperature and PPG heat-stroke detection systems already exist at research level, which makes patenting harder.
- **Wellness classification:** registering the device under the FDA's general wellness guidelines classes it as a wellness device rather than a clinical diagnostic tool, so the technical wording of the claims has to be specific.
- **Component level vs. system level:** the complete system is hard to patent, so protection is sought sub-system by sub-system. But these sub-systems are built from common components and modules, so the more finely we split the claims, the less unique each one becomes.

13.6 Data as an Asset

Longitudinal, anonymized multi-sensor user data collected in deployment is a proprietary dataset. With appropriate consent and anonymization, it can be used to improve the machine learning risk model and create a competitive advantage.

13.7 Freedom to Operate (FTO)

An FTO clearance review is planned before commercial launch, to confirm that the design does not infringe existing third-party patents in wearable physiological monitoring. The claim-by-claim comparison above feeds directly into this review.

14 Regulatory Considerations

14.1 Device Classification Strategy

VitaSync is positioned as a **general wellness product**: "a system that monitors and evaluates the risk of heat-stroke occurrence in occupational and athletic settings", rather than a diagnostic, preventive or mitigative medical device. This qualifies the product for the FDA's general wellness policy and avoids the Class II medical device pathway, which would otherwise require a full ISO 13485 Quality Management System, clinical trials, and a 12–24 month approval timeline before any revenue could be generated.

14.2 FDA General Wellness Requirements

To stay within the general wellness category, the device must meet six requirements:

1. non-invasive technology only;
2. no higher-risk interventions;
3. no disease diagnosis, treatment or prevention claims;
4. not a substitute for an FDA-cleared device;
5. no prompts to take specific clinical or medical actions based on diagnostic thresholds;
6. no clinically mimicking output values unless independently validated.

The device must also pass two tests:

- **Factor 1:** is the intended use limited to general wellness, or to lifestyle-linked disease-risk reduction where the link is well established?
- **Factor 2:** is it low risk, meaning non-invasive, with no implants, no harmful energy, and no meaningful risk if left unregulated?

As a result, the alerts shown on the app and fleet dashboard must indicate risk only. They must not give clinical advice or a diagnosis.

14.3 NMRA Registration Path (Sri Lanka)

The device is a wellness product under FDA rules, but it may still be treated as a medical device under the standards of the National Medicines Regulatory Authority (NMRA). The first step is therefore to confirm its classification with the NMRA. The expected path is:

1. **Clarify classification:** confirm with the NMRA whether the device is a medical device or a wellness device.
2. **Appoint a Marketing Authorization Holder (MAH):** a Sri Lanka-based legal entity is mandatory. It holds full regulatory accountability for registration, importation, sale, distribution and quality-failure handling. If VitaSync is not incorporated in Sri Lanka, these all run through the MAH and not through VitaSync directly.
3. **Sample import license:** needed if components or prototypes are imported for testing or registration.
4. **Foreign Manufacturing Site Approval (FMSA):** if any manufacturing takes place outside Sri Lanka, the site needs NMRA approval before product-level registration can proceed. One approval can support several later product registrations.
5. **Consolidated dossier:** since 5 August 2025, a single submission replaces what used to be sequential filings. It includes the device description, intended use, technical files, and English labeling for the Sri Lankan market.
6. **Internal technical evaluation** by NMRA staff.
7. **Medical Device Evaluation Committee (MDEC) review:** an expert panel evaluates quality, safety, effectiveness, need and cost. Existing FDA registration can be submitted as evidence to bypass the panel, unless the device is Class III.
8. **Product registration:** if approved, the NMRA issues a Registration Certificate.

9. **Import licensing:** a separate step after registration, required before the product can clear customs or be sold. Registration alone does not authorize sale.
10. **Advertising pre-approval:** promotional material needs prior written approval from the NMRA Advertising Evaluation Sub-Committee, which meets monthly and charges a processing fee.
11. **Post-market surveillance:** the MAH must report adverse events and quality issues, keep distribution records, and support recalls on an ongoing basis after launch.

Registration is expected to take at least about six months.

14.4 Other Safety Standards

Table 7 lists the safety standards that apply to the hardware and software.

Table 7: Additional safety standards

Standard	Scope
IEC 62133-2	Lithium battery safety for portable and wearable applications
ISO 10993-1 (with -5 cytotoxicity and -10 irritation/sensitization)	Biocompatibility of any component in sustained skin contact
IEC 62304	Software lifecycle for medical device software
IEC 60601	Electrical safety of medical electrical equipment

14.5 Labeling and Marketing Discipline

All product and marketing materials must consistently avoid language implying diagnosis, prevention or treatment of heat stroke. Otherwise the product could be reclassified as a regulated medical device (the "Diagnostic Reclassification Trap", identified as a killer risk in Section 16). The same wording rules apply to the NMRA advertising pre-approval step above.

15 Business Model

15.1 Selected Model: B2B Occupational Safety (Hardware-as-a-Service)

The highest-scoring business model from the concept screening matrix (Assignment 2) is retained as the primary go-to-market strategy. Vests are leased or sold to employers (agricultural estates, construction firms, and military users) with a rental or pilot tier and a purchase tier for scaled deployments. Managers get a free fleet-monitoring dashboard, and an optional paid compliance and reporting add-on is offered on top. This model aligns with employers' interest in reducing liability and improving labor efficiency during extreme heat.

15.2 Secondary and Deferred Models

- **B2B Elite Athletics** – rated "Revisit": a strong technical fit for a small, controlled user base, but the total addressable market is too small for it to be the primary commercial path at this stage. It remains the secondary segment.
- **B2C Consumer Wellness** – rated "No-Go": high customer acquisition cost and thin hardware margins make it unattractive as a near-term strategy.

15.3 Direct Competitors

- **Slate Safety BAND V2.0:** an armband that tracks heart rate, skin temperature, exertion and environmental factors. It is priced at $690 to buy, or $80 per day / $160 per week to rent. It uses edge-based cloud processing, is SOC-2 compliant, and offers API integrations for enterprise customers.
- **Kenzen:** a biosensor smart patch with an AI-driven platform for workplace safety.

15.4 Market Analysis

The main target sectors are occupational safety wear and athletic wear. The main industries are heavy machinery, energy, manufacturing, mining and heavy construction.

Table 8: Market size and growth

Report scope	2025/26 size	Forecast	CAGR
Worker heat-stress monitoring wearables (broadest; includes smart bands)	$0.57B (2025) to $0.64B (2026)	$1.04B by 2030	12.9%
Wearable temperature sensors (broadest adjacent category; all health and industrial)	$68.09B (2026)	$121.72B by 2031	12.34%; industrial/occupational safety is the fastest-growing vertical at 13.21%

Drivers of demand.

- **Regulation:** the proposed US federal OSHA heat standard would cover about 36 million workers if finalized. It is the largest single regulatory trigger in the target market, because a mandate would force employers to adopt monitoring rather than merely encourage it.
- **Climate:** by 2030, heat stress is projected to cause a 2.2% loss of total working hours worldwide, equivalent to 80 million full-time jobs and about $2.4 trillion in economic losses.

Market structure. No company dominates this growth-stage market. Most vendors are either instrument makers or wearable-band makers. VitaSync's uniqueness is that a widely available sensor-fusion model does not yet exist in the industry.

15.5 Go-to-Market Strategy

- **Primary segment:** occupational safety (construction, agriculture, military). **Secondary:** elite athletics.
- **Differentiation:** VitaSync measures the worker, using sensor fusion of GSR, PPG, skin temperature and ambient conditions, not just the environment.

1. **Phase 1 (0–12 months):** pilot with one or two Sri Lankan construction or agricultural employers, priced near cost to collect proof-of-concept data.
2. **Phase 2 (12–30 months):** expand to India and South Asia, and add the paid compliance-reporting layer once two or three reference customers exist.

3. **Phase 3 (30+ months):** enter the US and EU, timed to the finalization of the OSHA federal rule. The EU requires CE marking through a Notified Body.

Partnerships: insurers (claims reduction), OHS consultancies, and government and military procurement.

Revenue protection: because the dashboard is free, hardware margin must cover the backend costs (Firebase and ML hosting).

IP and regulatory alignment: patenting the sensor placement and fusion method protects VitaSync's position as competitors enter, and all marketing keeps to wellness-based language (Section 14).

15.6 Pricing and Revenue Model Summary

Table 9: Revenue model summary

Component	Model	Notes
Hardware (vest)	Rental/pilot tier and purchase tier	Margin must cover backend costs
Manager dashboard	Free	Drives adoption
Compliance/reporting add-on	Optional paid layer	Added in Phase 2
Data/analytics	Future tiered add-on	Builds on the paid add-on

16 Killer Risks and Mitigations

Table 10: Identified killer risks and mitigation strategies

Risk	Impact	Mitigation
PPG motion artifact	Corrupted heart-rate data during physical activity	Stable anatomical placement (C7 zone), co-located accelerometer for motion-referenced filtering, band-pass filtering (0.5–4 Hz).
Ambient light interference	Corrupted PPG signal	Extended housing lip and matte-black gasket to block external light leakage into the photodiode.
Single-marker false readings	Missed anhidrosis onset or false alarm from salt-crystal conductivity	Software cross-referencing of GSR trend against ambient humidity and rising skin temperature.
Regulatory reclassification	12–24 month delay, ISO 13485 burden	Strict "general wellness" marketing language, avoiding diagnostic/preventive claims.

17 Future Improvements

17.1 Hardware

- **Low-power MCU and battery life**: move from the ESP32 to a low-power BLE microcontroller (e.g. nRF52), as planned in the architecture, so the battery lasts a full shift. The phone acts as a gateway with offline buffering when there is no Wi-Fi. Duty-cycled excitation and sensor polling, and deep-sleep modes between measurement windows, reduce consumption further.
- **PCB and enclosure**: move from breadboard/dot-board prototyping to a custom PCB with proper analog/digital ground separation, reducing noise and improving manufacturability. Finalise the C7 patch enclosure in flexible TPU, with a rubber gasket against ambient light.
- **Water and sweat protection**: IP-rated, waterproof and sweat-resistant enclosure design for the electronics module, suitable for sustained outdoor and athletic use.
- **Electrode upgrade**: replace the crocodile-clip test connections with snap-type Ag/AgCl disposable electrodes and shielded leads, for better contact reliability and less motion artifact. Improve the GSR electrodes against sweat flooding and salt build-up.
- **Additional sensing**: add respiratory rate (recommended by the Medical Officer) and test green-LED PPG (MAX30105) for a better signal during movement.

17.2 Software and Machine Learning

- **On-device inference**: run a quantized version of the risk model on the device (TinyML) so prediction works without internet, e.g. on remote estates.
- **Personalisation**: personalised baselines, per-user fine-tuning, and acclimatisation tracking over days.
- **Core temperature estimation**: estimate core temperature from skin temperature, heart rate and environment, using heat-balance or Kalman-filter models.
- **Supervisor fleet dashboard**: show multiple workers, alert history and shift reports.
- **Multi-language mobile app support** for the target occupational markets.

17.3 Validation and Regulatory

- **Clinical validation**: a larger field study with a more diverse group of outdoor workers in Sri Lanka, compared against core-temperature references (rectal or ingestible pill) to validate the skin-temperature proxy and strengthen the evidence base.
- **Ethics and regulatory pathway**: obtain proper ethics approval. If medical claims are pursued later, an ISO 13485 quality management system and an NMRA/FDA Class II submission will be needed.

17.4 Mobile Application Regulatory Compliance

To ensure adherence to health software guidelines and data privacy standards, the mobile application incorporates the following regulatory and safety features:

- **Diagnostic Terminology Revision:** To classify the system as a preventative wellness tool rather than a medical diagnostic device, interface terminology was explicitly adjusted. References to "Heatstroke warning" were replaced with "Physiological Strain Monitor," and claims of "Heatstroke risk" were modified to assess the "likelihood of heat-stress."
- **User Interface and Medical Disclaimers:** All safety alerts were designed to be dismissible by the user to maintain user agency. Additionally, the application prominently displays an accuracy note alongside a mandatory disclaimer advising users to consult a medical professional rather than relying solely on the app.
- **Data Privacy and Explicit Consent:** Strict access controls are enforced so that workers can only view their own personal data, while all backend physiological metrics are collected and stored anonymously. The initial app launch features an onboarding screen to capture explicit user consent for data collection, detailing the exact duration of data retention.
- **Safety and Security Declarations:** The application includes a dedicated regulatory menu that formally declares the paired hardware's passed electrical safety certifications and specifies the data security encryption levels implemented across the cloud infrastructure.

18 Conclusion

This report has presented the complete engineering design and validation of the VitaSync wearable heat-stroke early-warning system, from the transistor-level analog GSR measurement circuit through to business model and regulatory strategy. The multi-sensor fusion approach – combining GSR, PPG, skin temperature, and ambient sensing – directly addresses the false-reading limitations identified in standalone-sensor approaches during concept screening, while the general-wellness regulatory positioning and Hardware-as-a-Service business model provide a practical path to market within the identified B2B occupational safety segment.

A Additional Figures

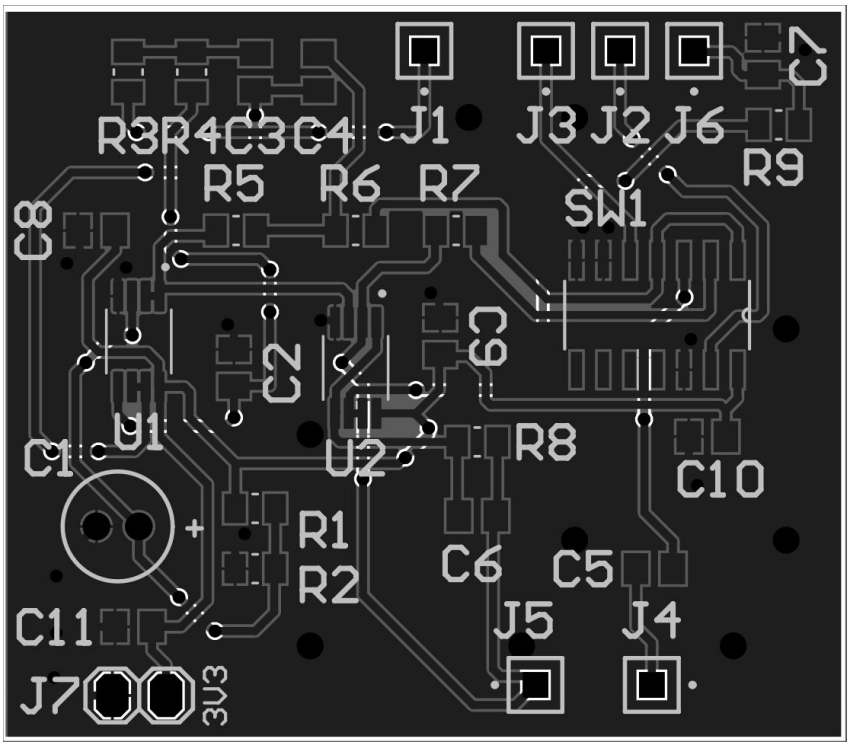


Figure 13: PCB layout of the GSR analog front end (Altium Designer), showing the OPA2333 op-amps (U1, U2), the 74HC4066 analog switch (SW1), passive components and the connector headers J1–J7.

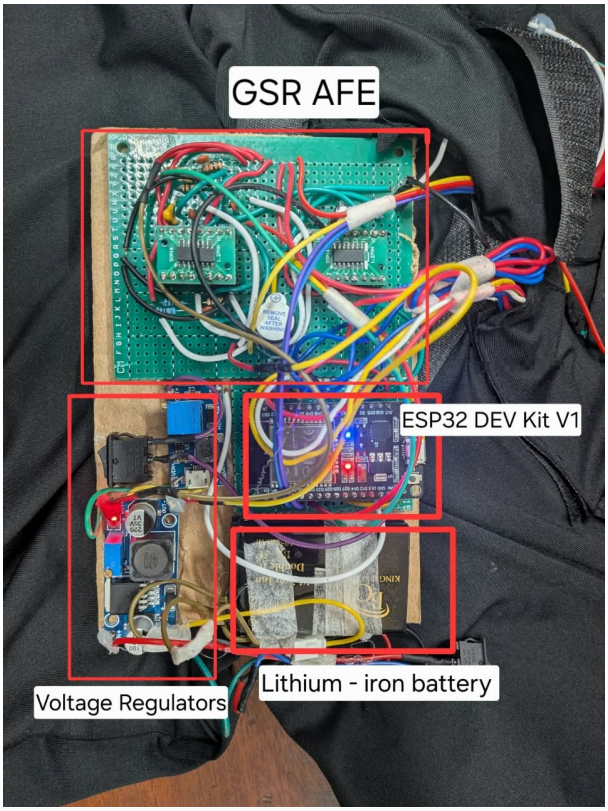


Figure 14: Control unit, GSR front end and power supply



(a) Back view



(b) Front view

Figure 15: VitaSync wearable vest prototype worn by a team member: (a) back view with the control unit pocket and sensor leads on the upper back, and (b) front view with the chest pocket.

B Github Repository

The complete source code, hardware schematics, and mobile application for the VitaSync project are available in the following GitHub repository:

https://github.com/ThennakoonUID/Early-Heat-Stroke-Warning-Wearable

B.1 Mathematical Models and Equations

1. Skin Temperature
The system computes the physical resistance of the NTC from the 12-bit ADC reading, then maps it to temperature in Celsius using the Beta parameter equation:[cite: 12]

$$R_{NTC} = R_{series} \cdot \left(\frac{4095}{raw_{ADC}} - 1 \right)$$

$$T_C = \left(\frac{1}{T_0} + \frac{1}{\beta} \ln \left(\frac{R_{NTC}}{R_0} \right) \right)^{-1} - 273.15$$

[8]

2. Environmental Metrics
SHT30 Temperature and Humidity Conversions
Transforms 16-bit raw sensor data into physical units:[cite: 12]

$$T_C = -45 + 175 \cdot \frac{raw_T}{65535}$$

$$RH\% = 100 \cdot \frac{raw_{RH}}{65535}$$

[9]

Heat Index (Rothfusz Regression)
Calculates the "feels like" temperature. It initially calculates a simple heat index:[cite: 12]

$$HI_{simple} = 0.5 \cdot (T_F + 61.0 + 1.2(T_F - 68.0) + 0.094 \cdot RH)$$

If the average of HI_{simple} and $T_F \geq 80°\text{F}$, it applies the full NWS 9-term polynomial regression:[cite: 12]

$$\begin{aligned} HI = &-42.379 + 2.049T_F + 10.143RH - 0.225T_F RH \\ &- 0.0068T_F^2 - 0.0548RH^2 + 0.0012T_F^2 RH \\ &+ 0.00085T_F RH^2 - 0.00000199T_F^2 \end{aligned}$$

[10]

3. Sweat Level Logarithmic Mapping
Maps skin conductance (g) to a 1–10 scale:[cite: 13]

$$Level = 1 + \text{round} \left(9 \cdot \frac{\ln(g/g_{min})}{\ln(g_{max}/g_{min})} \right)$$

[11, 12, 13]

4. Heart Rate & PPG Extrapolation[cite: 13]

$$HR_{max} = 208 - 0.7 \cdot Age$$

[14]

$$BPM = \frac{60 \cdot f_s \cdot (bin + \delta)}{N_{FFT}}$$

[15]

Perfusion Index (PI)
Calculates the AC-to-DC optical ratio:[cite: 13]

$$PI\% = 100 \cdot 2.83 \cdot \frac{AC_{RMS}}{DC_{EMA}}$$

5. Motion & IMU Math
Motion Feature Score
Uses vector magnitude RMS to determine the state of movement, independent of orientation:[cite: 14]

$$Score_{body} = \max\left(\frac{RMS_{acc_dynamic}}{Threshold_{acc}}, \frac{RMS_{gyro}}{Threshold_{gyro}}\right)$$

[16]

Dynamic Threshold Baseline Calibration
Sets individual motion gates after a 5-second stillness calibration:[cite: 14]

$$Threshold = \max(k_{mul} \cdot \mu, \mu + k_{std} \cdot \sigma, AbsoluteFloor)$$

[17, 18]

Thresholds & ML Model
[19, 20]

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